

LAB EXPERIMENT RECORD

Set 11: Product Category Analysis

Department	AIDS
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R
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Day	Wednesday

AIM

To analyze product category sales using R and Tableau by creating a funnel-style chart, sales table, pie chart, and an interactive Tableau dashboard.

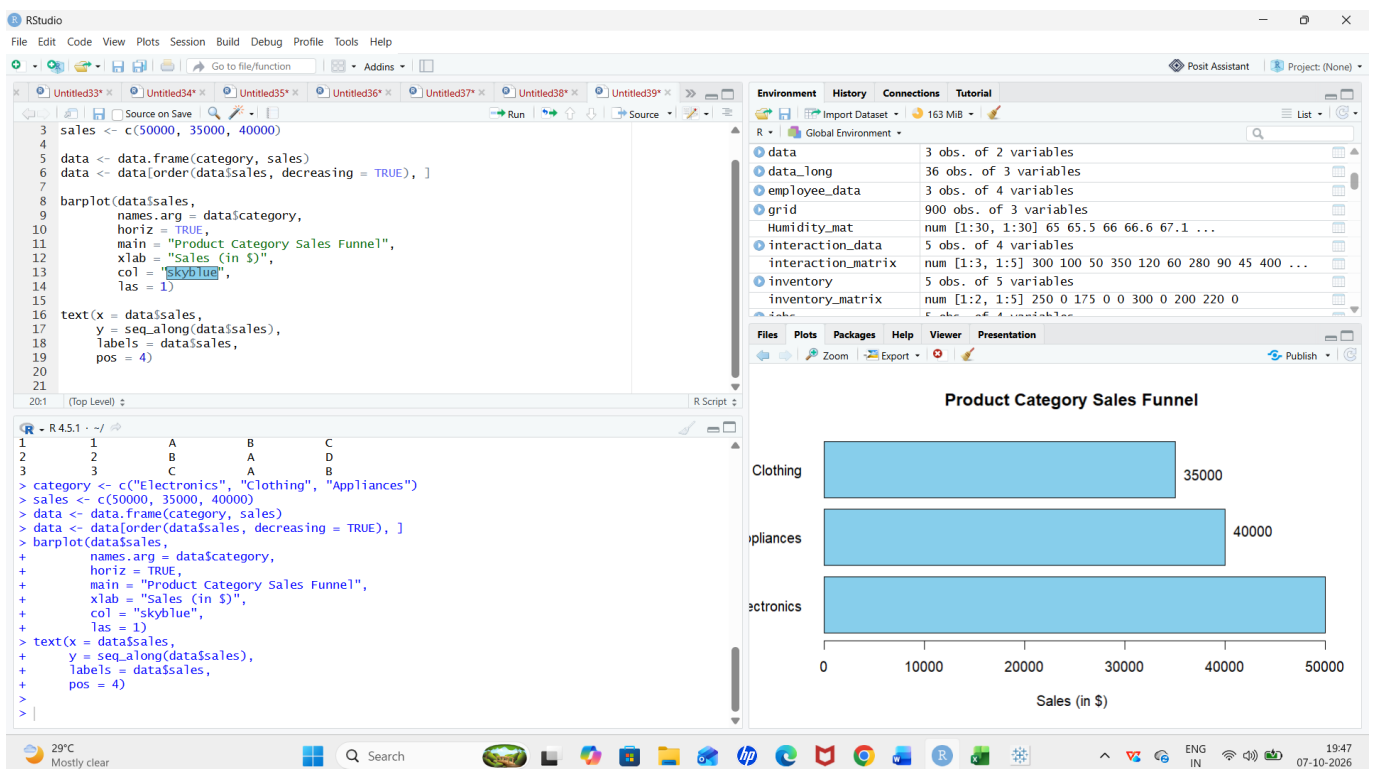
1. Generate a funnel chart to analyze the sales conversion process for each product category

```
category <- c("Electronics", "Clothing", "Appliances")
sales <- c(50000, 35000, 40000)
```

```
data <- data.frame(category, sales)
data <- data[order(data$sales, decreasing = TRUE), ]
```

```
barplot(data$sales,
        names.arg = data$category,
        horiz = TRUE,
        main = "Product Category Sales Funnel",
        xlab = "Sales (in $)",
        col = "skyblue",
        las = 1)
```

```
text(x = data$sales,
     y = seq_along(data$sales),
     labels = data$category,
     pos = 4)
```

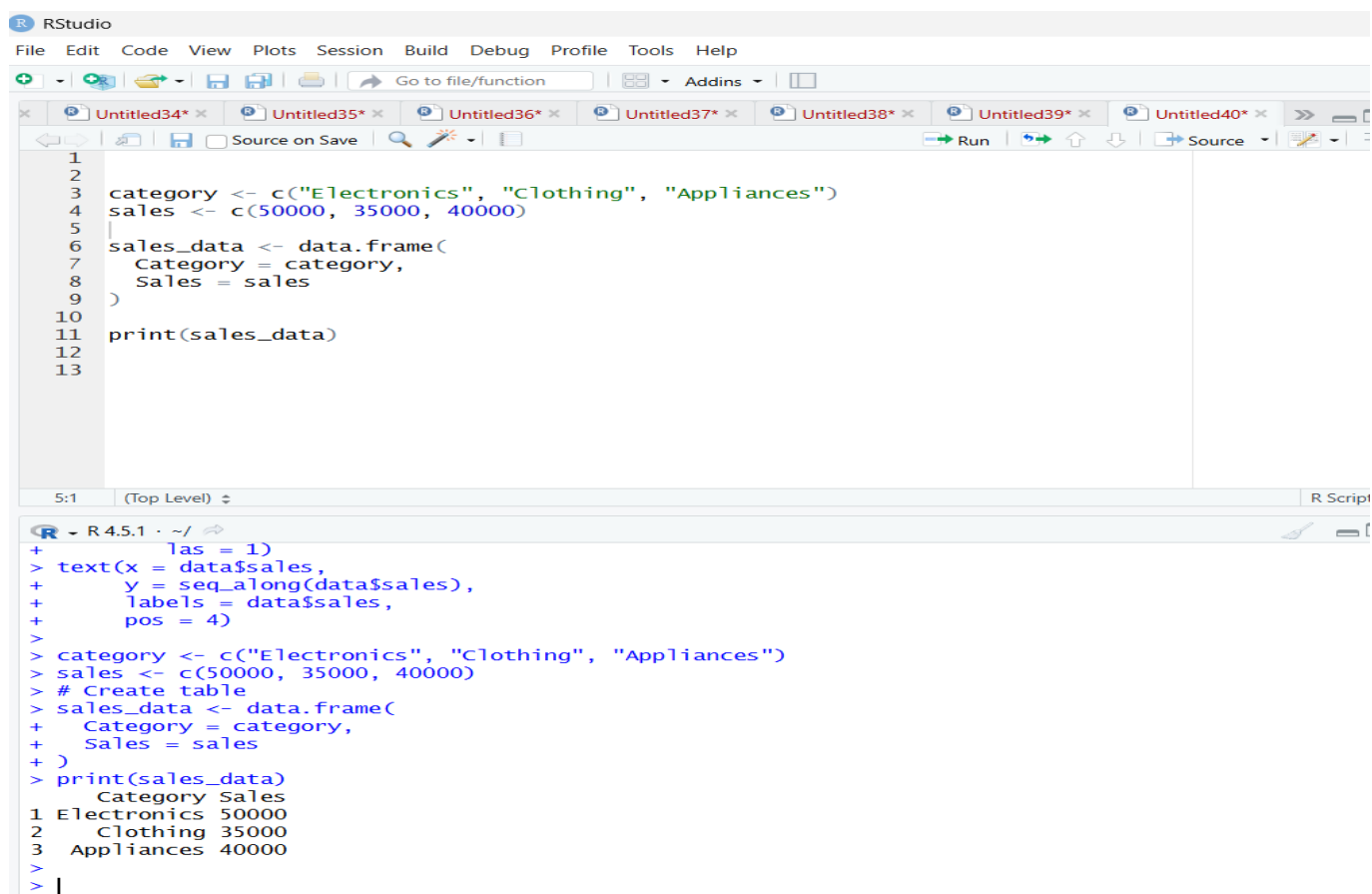


2. Build a table to display the sales data for each product category.

```
category <- c("Electronics", "Clothing", "Appliances")
sales <- c(50000, 35000, 40000)

sales_data <- data.frame(
  Category = category,
  Sales = sales
)

print(sales_data)
```



The screenshot shows the RStudio interface. The top pane displays the R script with line numbers 1 through 13. The bottom pane shows the console output, which includes the execution of the script and the resulting data frame.

```
1
2
3 category <- c("Electronics", "Clothing", "Appliances")
4 sales <- c(50000, 35000, 40000)
5
6 sales_data <- data.frame(
7   Category = category,
8   Sales = sales
9 )
10
11 print(sales_data)
12
13
```

```
R 4.5.1 ~ /
+ las = 1)
> text(x = data$Sales,
+      y = seq_along(data$Sales),
+      labels = data$Sales,
+      pos = 4)
>
> category <- c("Electronics", "Clothing", "Appliances")
> sales <- c(50000, 35000, 40000)
> # Create table
> sales_data <- data.frame(
+   Category = category,
+   Sales = sales
+ )
> print(sales_data)
  Category Sales
1 Electronics 50000
2  Clothing  35000
3 Appliances 40000
> |
```

4. . Create a pie chart to represent the distribution of sales across product categories. Include labels

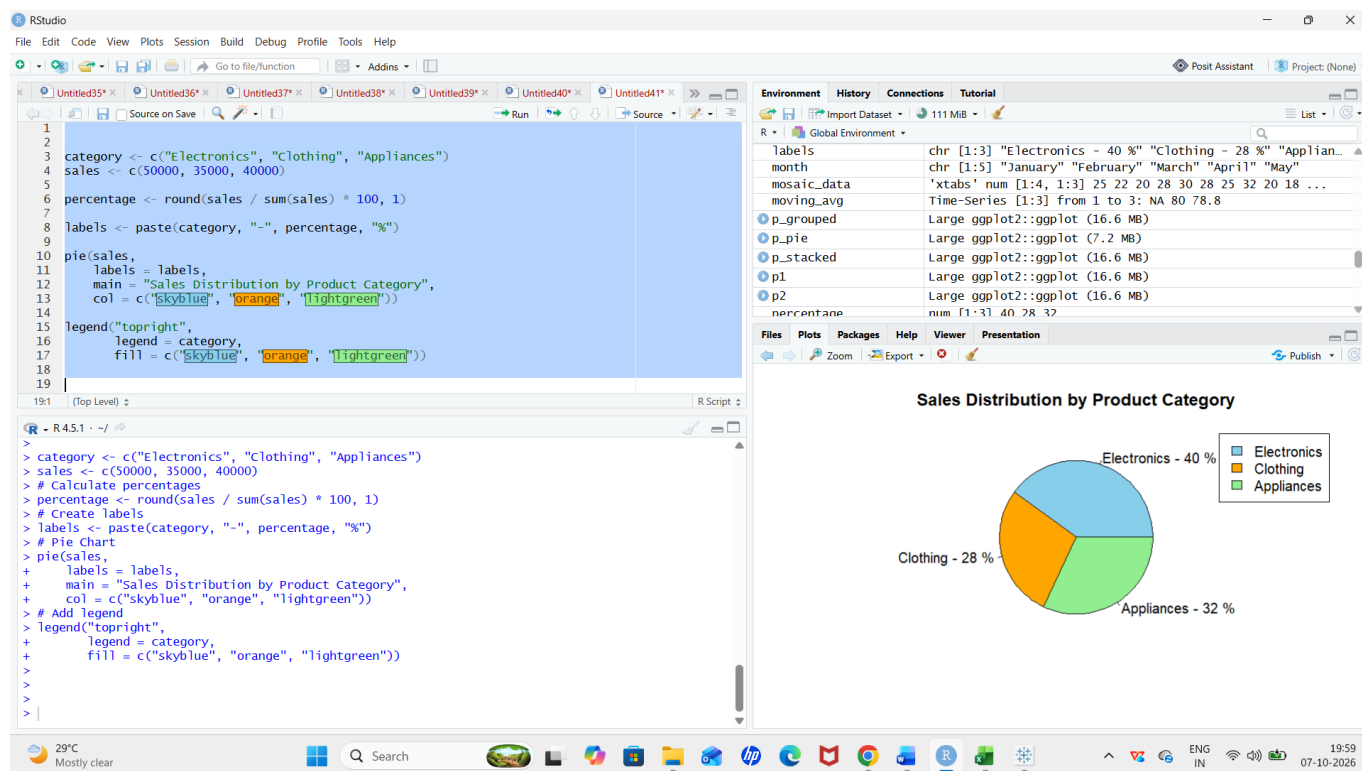
```
category <- c("Electronics", "Clothing", "Appliances")
sales <- c(50000, 35000, 40000)
```

```
percentage <- round(sales / sum(sales) * 100, 1)
```

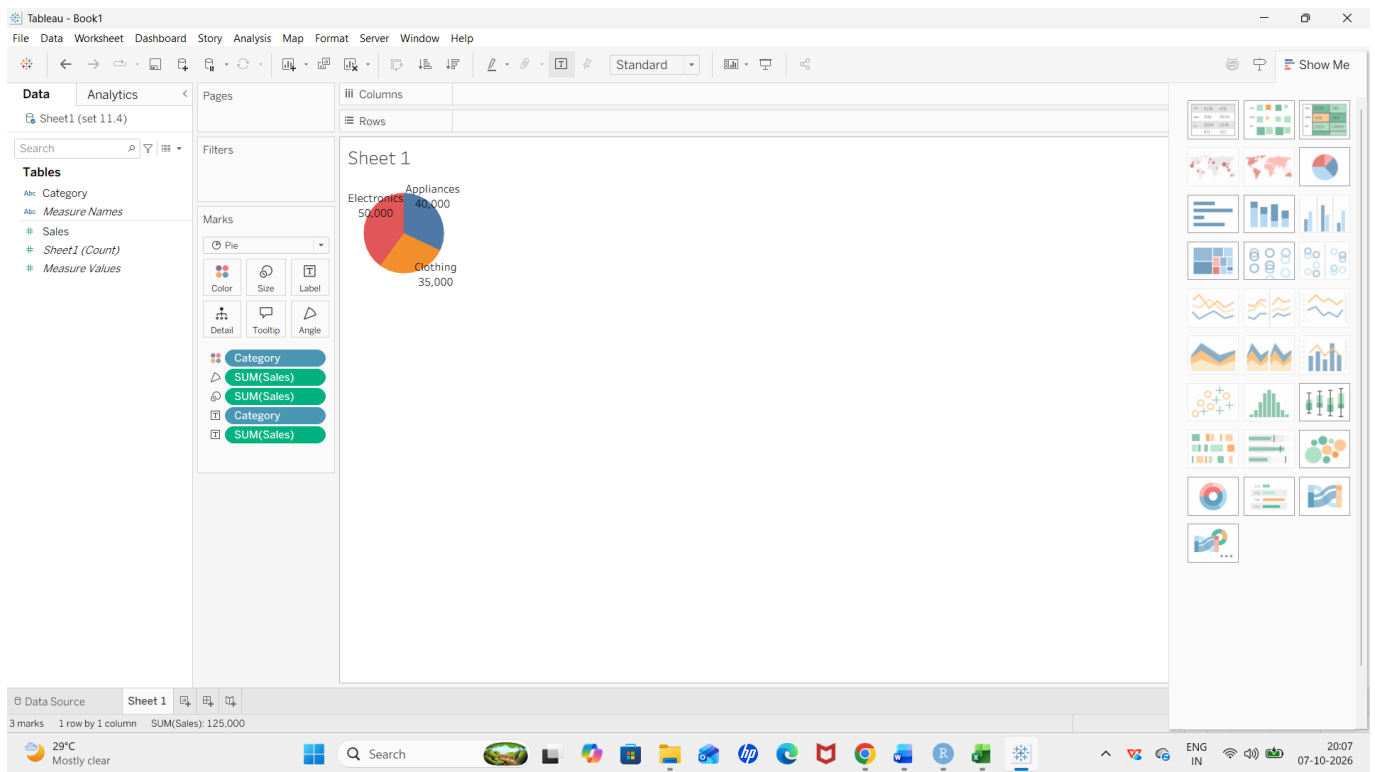
```
labels <- paste(category, "-", percentage, "%")
```

```
pie(sales,
    labels = labels,
    main = "Sales Distribution by Product Category",
    col = c("skyblue", "orange", "lightgreen"))
```

```
legend("topright",
    legend = category,
    fill = c("skyblue", "orange", "lightgreen"))
```



14.



PROCEDURE

1. Enter the product category and sales data into the dataset.
2. Create a **funnel-style chart** to compare sales across different product categories.
3. Create a **table** to display the sales value for each category.
4. Create a **pie chart** to visualize the distribution of sales among product categories.
5. Import the dataset into **Tableau**.
6. Create separate worksheets for the pie chart, funnel-style chart, and sales table.
7. Combine all three worksheets into a **Tableau dashboard**.
8. Add labels, titles, and filters to make the dashboard interactive.

RESULT

The product category sales data was successfully analyzed using R and Tableau. The pie chart shows the sales distribution, the funnel-style chart compares category sales, and the table displays the exact sales values. An interactive Tableau dashboard was created by combining all three visualizations.

